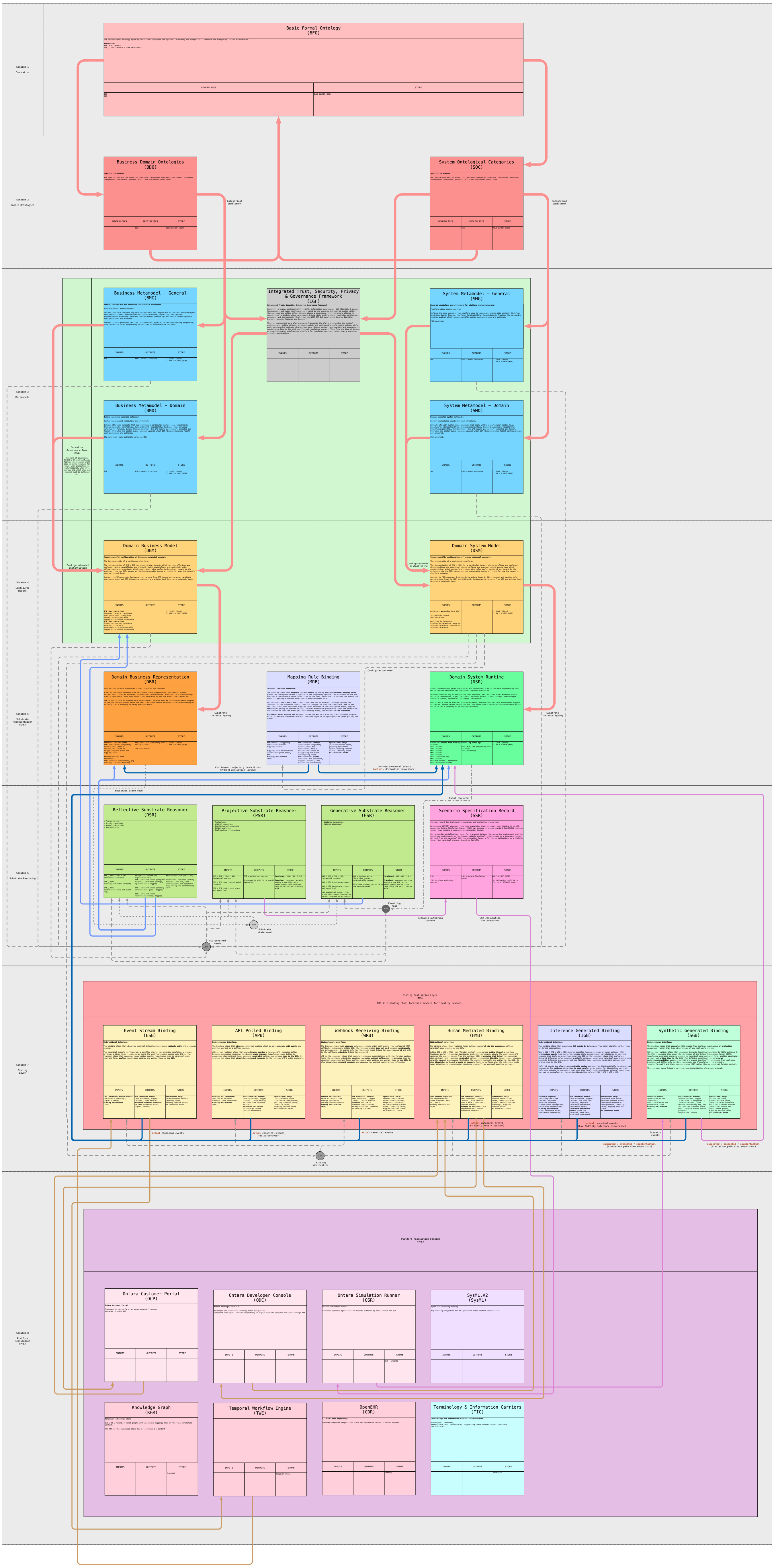

Categorical commitment
Categorical commitment
Formalism Governance Zone (FGZ)
Configured-model instantiation
Configured-model instantiation
Substrate instance typing
Substrate instance typing
Configuration read
Continuant trajectory transitions (PROV-O derivation-linked)
Derived canonical events (actual, derivation provenance)
Substrate state read
Event log read
Event log read
JOIN
Substrate state read
FGZ-governed reads
Scenario authoring content
SSR consumption for execution
actual canonical events
actual canonical events (delta-derived)
actual canonical events
actual canonical events (+ user / role / session)
actual canonical events (Low fidelity inference provenance)
Scenario events
simulated / projected / counterfactual (Simulation path also shows this)
simulated / projected / counterfactual (Simulation path also shows this)
Binding declaration

Stratum 1

Foundation

Basic Formal Ontology (BFO)

The shared upper ontology spanning both sides (business and system), providing the categorical framework for everything in the architecture.

Foundation:
BFO 2020 (upper)
CCO / IAO / PROV-O / OGMS (mid-level)

GENERALISES	STORE
BDO SOC	OWL2 DL/RDF (KGR)

Stratum 2

Domain Ontologies

Business Domain Ontologies (BDO)

Specific to domains.

BDO specialises BFO. It draws its top-level categories from BFO (continuant, occurrent, independent continuant, process, etc.) and subclasses under them.

GENERALISES	SPECIALISES	STORE
	FGZ	OWL2 DL/RDF (KGR)

System Ontological Categories (SOC)

Specific to domains.

SOC specialises BFO. It draws its top-level categories from BFO (continuant, occurrent, independent continuant, process, etc.) and subclasses under them.

GENERALISES	SPECIALISES	STORE
	FGZ	OWL2 DL/RDF (KGR)

Stratum 3

Metamodels

Formalism Governance Zone (FGZ)

The zone of governance forOWL 2 DL and SysML.v2. Applies rules about which side is authoritative for what, when projection is bidirectional, when it's one-way and which side new content must be authored on.

Business Metamodel – General (BMG)

General vocabulary and structure for service businesses

Platform-wide, domain-neutral.

Defines the core concepts any service business has, regardless of sector: ServiceSubject, ServiceParticipant, ServiceOffering, ServiceEpisode, Capability, Obligation, StakeholderRelationship. Provides the metamodel lexicon against which tenant-specific configurations are authored.

Content is FGZ-governed: OWL 2 DL is canonical, SysML v2 is the engineering projection, with authority rules determining which side is authoritative for what.

INPUTS	OUTPUTS	STORE
BDO	RSM – model structure	1. SysML (Repo) 2. OWL2 DL/RDF (KGR)

Integrated Trust, Security, Privacy & Governance Framework (IGF)

Integrated Trust, Security, Privacy & Governance Framework

Security, privacy, confidentiality, GDPR, information governance, IAM (Identity & Access Management), and cyber resilience is treated as one coordinated control system rather than as separate policy silos. Ontara adopts a governance-first structure aligned to modern cyber practice, where governance drives risk, architecture, controls, monitoring, response, and improvement. (Much like the NIST CSF 2.0 model with Govern, Identify, Protect, Detect, Respond, and Recover).

This is implemented as a platform meta-framework: the platform provides the control architecture, policy objects, evidence model, and configurable enforcement points, while each implementing business chooses how much rigour, review, segregation, and assurance is needed according to its risk profile and regulatory context. This fits the idea of Ontara as a multi-tenant, model-driven platform for regulated services rather than a one-size-fits-all application.

INPUTS	OUTPUTS	STORE

System Metamodel – General (SMG)

General vocabulary and structure for platform system behaviour

Platform-wide, domain-neutral.

Defines the core concepts the platform uses to represent system-side content: Workflow, Activity, Event, Binding, Surface, ConstraintRule, BoundedAgent. Provides the metamodel lexicon against which tenant-specific system configurations are authored.

FGZ-governed.

INPUTS	OUTPUTS	STORE
SOC	RSM – model structure	1. SysML (Repo) 2. OWL2 DL/RDF (KGR)

Business Metamodel – Domain (BMD)

Domain-specific business metamodel

Sector-specialised vocabulary and structure.

Extends BMG with concepts that apply within a particular sector (e.g. healthcare: ClinicalEpisode, CarePathway, ConsentRecord; hospitality: Booking, Tab, Service; veterinary: Patient, Owner, ClinicalCourse). One BMD module per sector, selected per tenant. Provides the sector-aware lexicon against which DBM (Domain Business Model) configurations are authored.

FGZ-governed, same authority rules as BMG

INPUTS	OUTPUTS	STORE
BDO	RSM – model structure	1. SysML (Repo) 2. OWL2 DL/RDF (KGR)

System Metamodel – Domain (SMD)

Domain-specific system metamodel

Sector-specialised vocabulary and structure.

Extends SMG with system-side concepts that apply within a particular sector (e.g. healthcare: ClinicalWorkflow, ConsultationSurface, PrescribingAction; hospitality: OrderPickingWorkflow, TillSurface). One SMD module per sector, selected per tenant. Provides the sector-aware lexicon against which DSM (Domain System Model) configurations are authored.

FGZ-governed.

INPUTS	OUTPUTS	STORE
SOC	RSM – model structure	1. SysML (Repo) 2. OWL2 DL/RDF (KGR)

Stratum 4

Configured Models

Domain Business Model (DBM)

Tenant-specific configuration of business metamodel concepts

The business-side of a configured platform.

The instantiation of BMG + BMD for a particular tenant: which service offerings are declared, which capabilities are claimed, which stakeholders are modelled, which obligations are recognised, which constraint rules apply. Authored per tenant by the architect via the ODC; serves as the business-side source of truth for what the tenant's service is and does.

Content is FGZ-governed. Derived-write targets from RSR (computed targets, candidate configurations) and GSR (directive content) are written back here with epistemic tags.

INPUTS	OUTPUTS	STORE
RSR (Derived write) Computed targets, candidate configurations, trajectory targets – epistemically tagged with PROV-O provenance **GSR (Derived write)** Directive content (guidance artefacts, valence assessments) – epistemically tagged with PROV-O provenance		1. SysML (Repo) 2. OWL2 DL/RDF (KGR)

Domain System Model (DSM)

Tenant-specific configuration of system metamodel concepts

The system-side of a configured platform.

The instantiation of SMG + SMD for a particular tenant: which workflows are declared, which bindings are specified, which surfaces are exposed, which agents have which capabilities, which system-level constraint rules apply. Authored per tenant by the architect via the ODC; serves as the system-side source of truth for how the tenant's platform behaves.

Content is FGZ-governed. Binding declarations (read by BRL classes) and mapping rule declarations (read by MRB) are DSM-held. Derived-write targets from RSR are written back here with epistemic tags.

INPUTS	OUTPUTS	STORE
Architect authoring (via ODC) System-side tenant configuration: workflow declarations, binding declarations, mapping rule declarations, constraint rule declarations		1. SysML (Repo) 2. OWL2 DL/RDF (KGR)

Stratum 5

Substrate Representation (SRS)

Domain Business Representation (DBR)

Data on the service activities – the 'state of the business'

A set of business entities with versioned state trajectories. Customers, orders, obligations, clinical episodes, stakeholder relationships. Each entity's state-at-any-time is queryable, with each transition annotated by the DSR event that caused it.

DBR is not a site of runtime rule enforcement, because runtime rule enforcement happens at the BRL before writes reach the DBR. The store itself enforces structural/ontological validity (as a property of being KGR-resident).

INPUTS	OUTPUTS	STORE
Canonical writes from: MRB: continuant trajectory transitions (PROV-O derivation-linked to triggering DSR event and mapping rule) **Derived writes from reasoners:** RSR: anomaly annotations, gap findings (actual-derived)	RSR, PSR, GSR (reasoning over entity state) IGB (evidence)	1. SysML (Repo) 2. OWL2 DL/RDF (KGR)

Mapping Rule Binding (MRB)

Internal reaction interface:

The binding class that **responds to DSR events** by firing **configured-model mapping rules**, producing consequent writes – typically DBR trajectory updates (an occurrent in the DSR causing a continuant's state transition in the DBR), occasionally further DSR events (an event triggering a derived event per a model-declared rule).

Unlike ESB / APB / WRB / HMB / IGB / SGB, MRB has no external foreign system; its "source" is the substrate itself, and its "target" is also the substrate. MRB is the realiser class that evaluates mapping rules declared in the configured model, applies **constraint** gating to derived writes, records derivation provenance (this DBR transition was caused by this DSR event per this mapping rule), and **writes to the substrate**.

Placement note: Whether MRB belongs inside the BRL as a binding class (current diagram) or as a separate substrate-internal reaction layer is an open question (S234 §12 Q4; see Q-MRB-1).

INPUTS	OUTPUTS	STORE
DSR events (triggering substrate-internal mapping rules) Mapping rule declarations (from configured model, KGR) **Binding declaration** (KGR)	**DBR canonical writes:** (continuant trajectory transitions; BFO continuant, PROV-O derivation-linked to triggering DSR event and mapping rule) **DSR canonical events** (derived; BFO occurrent, tagged `actual`, with derivation provenance)	**Operational only:** rule evaluation state, pending-derivation queue, mapping version state, identity cache. **No canonical truth.**

Domain System Runtime (DSR)

A multi-dimensional graph capture of all operational simulation data representing real world runtime execution and any other computed simulation.

An event-sourced log of everything that happened, real or simulated: Workflow starts, activity completions, signal arrivals, decision points, timer firings. Time-stamped, causally linked, epistemically tagged, replayable.

DSR is not a site of runtime rule enforcement, because runtime rule enforcement happens at the BRL before writes reach the DSR. The store itself enforces structural/ontological validity (as a property of being KGR-resident).

INPUTS	OUTPUTS	STORE
Canonical events from binding classes: ESB: actual APB: actual WRB: actual HMB: actual IGB: actual SGB: simulated etc. MRB: actual **Derived writes – reasoners:** RSR: detection events	**Event log reads by:** RSR, PSR, GSR (reasoning over history) IGB (evidence)	1. SysML (Repo) 2. OWL2 DL/RDF (KGR)

Stratum 6

Substrate Reasoning

Reflective Substrate Reasoner (RSR)

- Trajectories
- History analysis
- Anomaly detection
- Gap analysis

INPUTS	OUTPUTS	STORE
BMG / BMD / SMG / SMD (metamodel content) DBM / DSM (configured model content) DBR / DSR (substrate state and event log)	**Analytical output** (to experience-API) DBM – derived write (computed targets, candidate configs, epistemically tagged) DSR – derived write (anomaly annotations, gaps – tagged) DSR – derived write (detection events, tagged)	**Persistent:** KGR (OWL 2 DL) **Transient:** reasoner working memory – one KGR, with named graphs and epistemic tags doing the partitioning work

Projective Substrate Reasoner (PSR)

- Projections
- What-if scenarios
- Counterfactual analysis
- Option analysis
- Goal-seeking / solutions

INPUTS	OUTPUTS	STORE
BMG / BMD / SMG / SMD (metamodel content) DBM / DSM (configured model content) DBR / DSR (substrate state and event log)	SSR – authoring content (consumed by OSR for scenario execution)	**Persistent:** KGR (OWL 2 DL) **Transient:** reasoner working memory – one KGR, with named graphs and epistemic tags doing the partitioning work

Generative Substrate Reasoner (GSR)

- Guidance generation
- Valence assessment

INPUTS	OUTPUTS	STORE
BMG / BMD / SMG / SMD (metamodel content) DBM / DSM (configured model) DBR / DSR (substrate state and event log) RSR analytical output, PSR projection output (reasoning output consumed as evidence)	DBM – derived write (directive content, epistemically tagged) Directive content to surfaces via experience-API	**Persistent:** KGR (OWL 2 DL) **Transient:** reasoner working memory – one KGR, with named graphs and epistemic tags doing the partitioning work

Scenario Specification Record (SSR)

Storage record for individual simulation and projection scenarios.

References DBM/DSM versions, starting snapshots, input streams, etc. Keeping it in OWL means the Ontara Simulation Runner (OSR) can consume it using standard OWL/SPARQL tooling rather than needing a separate serialisation format.

For a non-OWL serialisation, e.g. for transport between the authoring environment and the execution environment, or for human-readable archival – that might be a secondary format derived from the canonical OWL representation (e.g., a Turtle serialisation, or a JSON-LD form). But canonical storage should be OWL/KGR.

INPUTS	OUTPUTS	STORE
PSR SSR receives authoring content.	OSR – Ontara Simulation Runner	OWL2 DL/RDF (KGR) Serialisation could be in Turtle or JSON-LD form.

Stratum 7

Binding Layer

Binding Realisation Layer (BRL)

MRB is a binding class located elsewhere for locality reasons.

Event Stream Binding (ESB)

Bidirectional interface:

The binding class that **observes** external infrastructure which **natively emits** state-change events.

The canonical example is Temporal: a workflow starts, an activity completes, a signal arrives, a timer fires – each is an event the workflow engine pushes out. ESB is the realiser class that **receives** those native events, **translates** them to canonical edge contract form, **applies constraint** gating, and **writes them to the DSR.**

INPUTS	OUTPUTS	STORE
TWE (workflow) native events: (workflow / activity / signal / timer) **Binding declaration** (KGR)	**DSR canonical events:** (BFO occurrent, tagged `actual`, with PROV-O) **Outbound workflow commands to TWE:** (workflow start, signal, cancel)	**Operational only:** reconciliation cursors, identity cache, mapping version state. No canonical truth.

API Polled Binding (APB)

Bidirectional interface:

The binding class that **observes** external systems which **do not natively emit events** and must be queried on a polling cadence.

APB is the realiser class that **periodically queries** a foreign API, computes deltas between successive responses to **detect state changes, translates** those deltas to canonical edge-contract form, applies **constraint** gating, and **writes them to the DSR.** It is also the class that **dispatches outbound commands** to the foreign API's write endpoints.

INPUTS	OUTPUTS	STORE
Foreign API responses: (polled on declared cadence, read endpoints) **Binding declaration** (KGR)	**DSR canonical events:** (BFO occurrent, tagged `actual`, with PROV-O; derived from response deltas) **Outbound API calls:** to foreign system (write endpoints)	**Operational only:** last-response cache (for delta computation), polling schedule state, identity cache, mapping version state. No canonical truth.

Webhook Receiving Binding (WRB)

Bidirectional interface:

The binding class that **observes** external systems which emit events via configured HTTP callbacks (webhooks). Unlike ESB, the foreign system **does not push events continuously** and autonomously; it pushes only for the event types Ontara has **pre-registered** and only to the **callback endpoints** Ontara has declared.

WRB is the realiser class that registers webhook subscriptions with the foreign system, hosts the callback endpoints, **receives incoming webhook deliveries, translates** them to canonical edge-contract form, applies **constraint** gating, and **writes them to the DSR.** It also **dispatches outbound commands** and **manages** the webhook registration lifecycle.

INPUTS	OUTPUTS	STORE
Webhook deliveries: (HTTP callbacks from foreign system on pre-registered events) **Binding declaration** (KGR)	**DSR canonical events:** (BFO occurrent, tagged `actual`, with PROV-O) **Outbound API calls** (webhook registration, re-registration, commands to foreign system)	**Operational only:** webhook registration state (subscriptions, endpoints, secrets), delivery deduplication cache, identity cache, mapping version state. No canonical truth.

Human Mediated Binding (HMB)

Bidirectional interface:

The binding class that realises human actions **captured via the experience-API** as canonical-edge events in the DSR.

Unlike ESB / APB / WRB, the "foreign system" is **a person acting through a surface** (customer portal, clinician dashboard, architect workspace, etc.). The experience-API captures the user's intent from the surface; HMB **translates that intent** to canonical edge-contract form, applies **constraint** gating, reconciles **identity** (user, role, session, tenant), **records provenance** (including the capture context), and **writes to the DSR.** It also **dispatches outbound prompts or requests** back to surfaces when the substrate needs human attention (a consultation requiring sign-off, an approval awaiting action).

INPUTS	OUTPUTS	STORE
User intents captured via experience-API (from surfaces: OCP, ODC, etc.) Binding declaration (KGR)	**DSR canonical events:** (BFO occurrent, tagged `actual`, with PROV-O; user / role / session / capture context) **Prompts to surfaces** (via experience-API; human attention requests)	**Operational only:** session correlation state, identity cache (user → Ontara runtime identity), mapping version state. No canonical truth.

Inference Generated Binding (IGB)

Bidirectional interface:

The binding class that **generates DSR events by inference** from other signals, rather than by direct observation.

Where ESB / APB / WRB / HMB observe specific foreign systems or human actions, IGB **synthesises events** from patterns, steady-state assumptions, correlations, or derived signals that imply an event has occurred. IGB is the realiser class that evaluates declared inference rules against available evidence, generates canonical-edge events with explicit inference provenance and low fidelity tags, applies constraint gating, and writes them to the DSR.

Inferred events are **always epistemically marked** and must be distinguishable to downstream consumers. The **outbound direction is used rarely**, principally for dispatching derived-inference outputs to consumers that need them (analytical pipelines, external reporting) but has no equivalent of the action-dispatching role of ESB / APB / WRB / HMB.

INPUTS	OUTPUTS	STORE
Evidence signals: (existing DSR / DBR content, declared baseline patterns, steady-state assumptions, correlated observations) **Binding declaration** (KGR; inference rules, confidence thresholds)	**DSR canonical events:** (BFO occurrent, tagged `actual` but with low fidelity, explicit inference provenance, confidence level) **Inference provenance events** (what was inferred, from what, with what confidence)	**Operational only:** inference rule state, baseline pattern cache, confidence-threshold configuration, identity cache, mapping version state. **No canonical truth.**

Synthetic Generated Binding (SGB)

Bidirectional interface:

The binding class that **generates DSR events** from declared **simulation or projection scenarios**, rather than from observation of any real-world system.

SGB is the realiser class that consumes Scenario Specification Records (SSRs authored by the PSR), executes them under the direction of the Ontara Simulation Runner (OSR), **translates** generated scenario events to canonical edge-contract form, applies **constraint** gating, and **writes them to the DSR.** The defining commitment is **structural indistinguishability:** events from SGB are shaped identically to events from real-mode bindings and differ only in their epistemic tag (`simulated`, `projected`, `counterfactual`) and their source system (OSR rather than an external foreign system).

This is what makes Ontara's unity-across-instantiation claim operational.

INPUTS	OUTPUTS	STORE
Scenario events (generated by OSR from SSRs) SSRs (referenced for provenance, KGR) **Binding declaration** (KGR)	**DSR canonical events:** (BFO occurrent, tagged `simulated` / `projected` / `counterfactual`, with PROV-O referencing SSR, run ID, and the scenario clock) Run-lifecycle events (start, progress, completion, fault)	**Operational only:** active run state, scenario-clock state, identity cache (scenario entities → Ontara runtime identities within the run's named graph), mapping version state. **No canonical truth.**

Stratum 8

Platform Realisation (PRS)

Platform Realisation Stratum (PRS)

Ontara Customer Portal (OCP)

Ontara Customer Portal

Customer-facing surface; an experience-API consumer mediated through HMB

INPUTS	OUTPUTS	STORE

Ontara Developer Console (ODC)

Ontara Developer Console

Developer and architect surface; model navigation, component catalogue, runtime inspection; an experience-API consumer mediated through HMB

INPUTS	OUTPUTS	STORE

Ontara Simulation Runner (OSR)

Ontara Simulation Runner

Executes Scenario Specification Records authored by PSR; source for SGB

INPUTS	OUTPUTS	STORE
		KGR – GraphDB

SysML.V2 (SysML)

SysML v2 authoring tooling

Engineering projection for FGZ-governed model content (strata 2–4)

INPUTS	OUTPUTS	STORE

Knowledge Graph (KGR)

Canonical substrate store

OWL 2 DL / SPARQL / named graphs with epistemic tagging; home of the full stratified content.

The KGR is the canonical store for all stratum 1–5 content

INPUTS	OUTPUTS	STORE
		GraphDB

Temporal Workflow Engine (TWE)

INPUTS	OUTPUTS	STORE
		Temporal store

OpenEHR (CDR)

Clinical data repository

openEHR-compliant composition store for healthcare tenant clinical records

INPUTS	OUTPUTS	STORE
		EHRBase

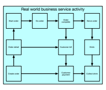
Terminology & Information Carriers (TIC)

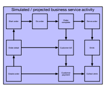
Terminology and information-carrier infrastructure

Archetypes, templates, SNOMED/LOINC/etc. vocabularies, supporting coded content across substrate and surfaces

INPUTS	OUTPUTS	STORE
		EHRBase?

Ontara Platform
v3.4.1
30/04/2026


Real world business service activity


Simulated / projected business service activity